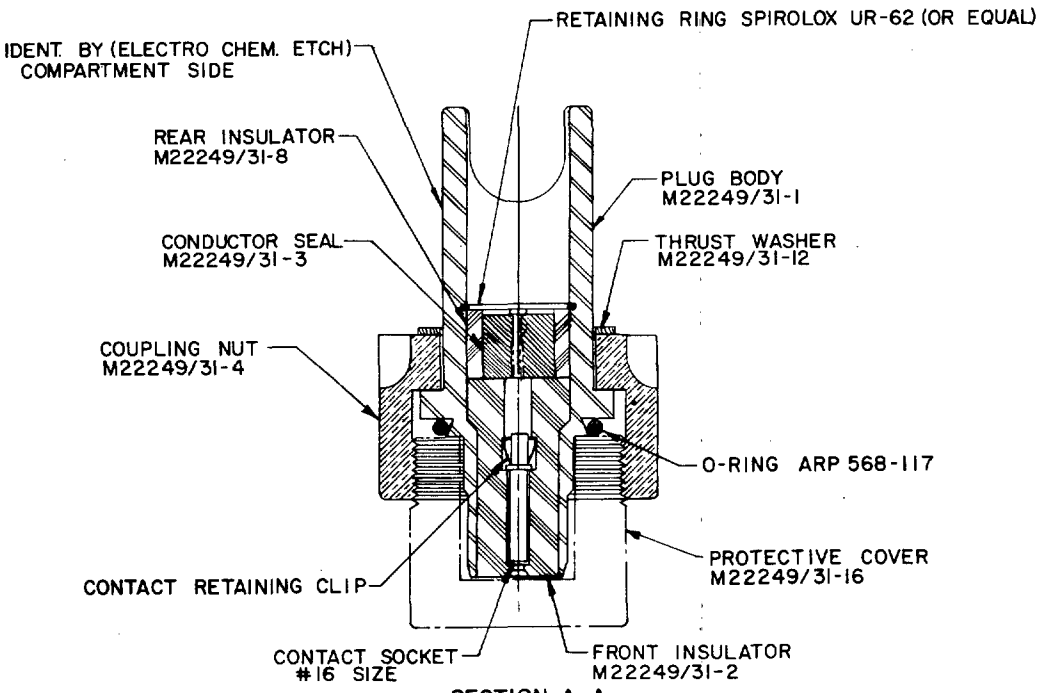
MILITARY SPECIFICATION SHEET

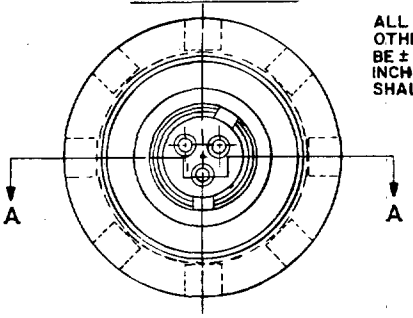
CONNECTOR SETS, PLUG ASSEMBLY, 3-PIN, COMPARTMENT SIDE

(A) This specification sheet is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connector sets described herein shall consist of this document and the latest issue of MIL-C-22249(SH).



SECTION A-A



ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE ± 1/64 INCH FRACTIONS AND ± 0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE SHALL BE ± 1/2 DEGREE.

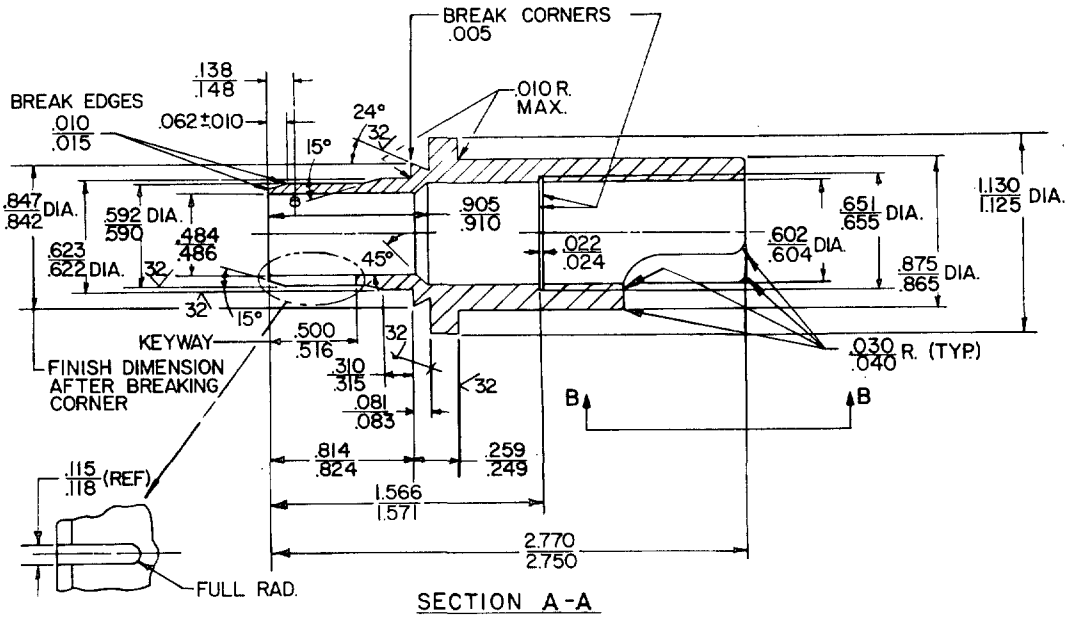
NOTE:
PLUG TO BE SHIPPED IN ASSEMBLED CONDITION EXCEPT FOR CONTACTS.

(A) DENOTES CHANGE

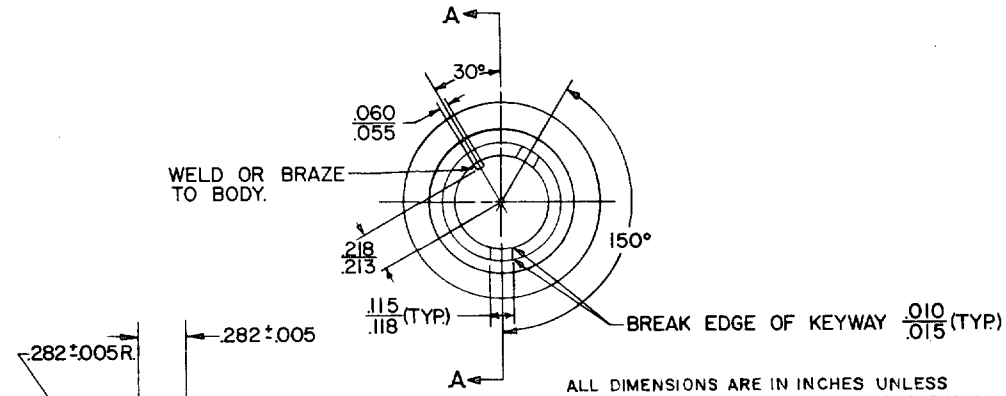
FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:39Z
Check the source to verify that this is the current version before use.

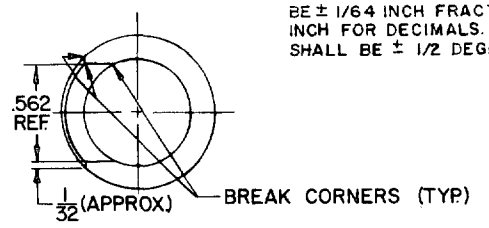
PLUG BODY
M22249/31-1



SECTION A-A



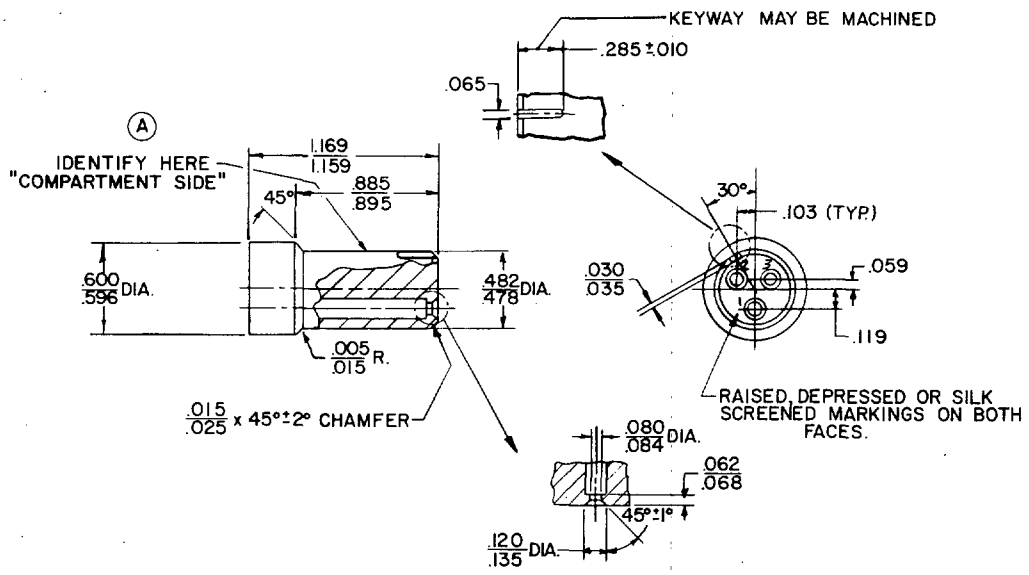
ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE ± 1/64 INCH FRACTIONS AND ± 0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE SHALL BE ± 1/2 DEGREE.



VIEW B-B

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:39Z
Check the source to verify that this is the current version before use.

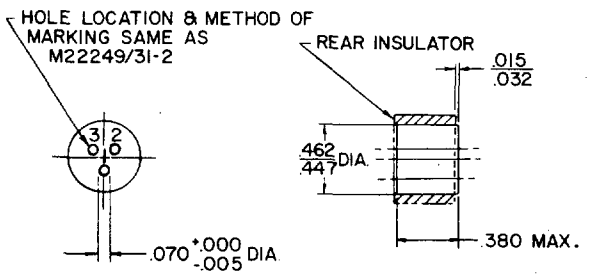
FRONT INSULATOR
M22249/31-2



INSULATOR MATERIAL: TO BE MOLDED FROM DIALLYL PHTHALATE IN ACCORDANCE WITH MIL-C-22249.

NOTE:
SOCKET CONTACTS TO BE FURNISHED WITH INSULATOR BUT NOT IN PLACE.
SILICONE GROMMET NOT ATTACHED TO FRONT INSULATOR.

CONDUCTOR SEAL GROMMET
M22249/31-3

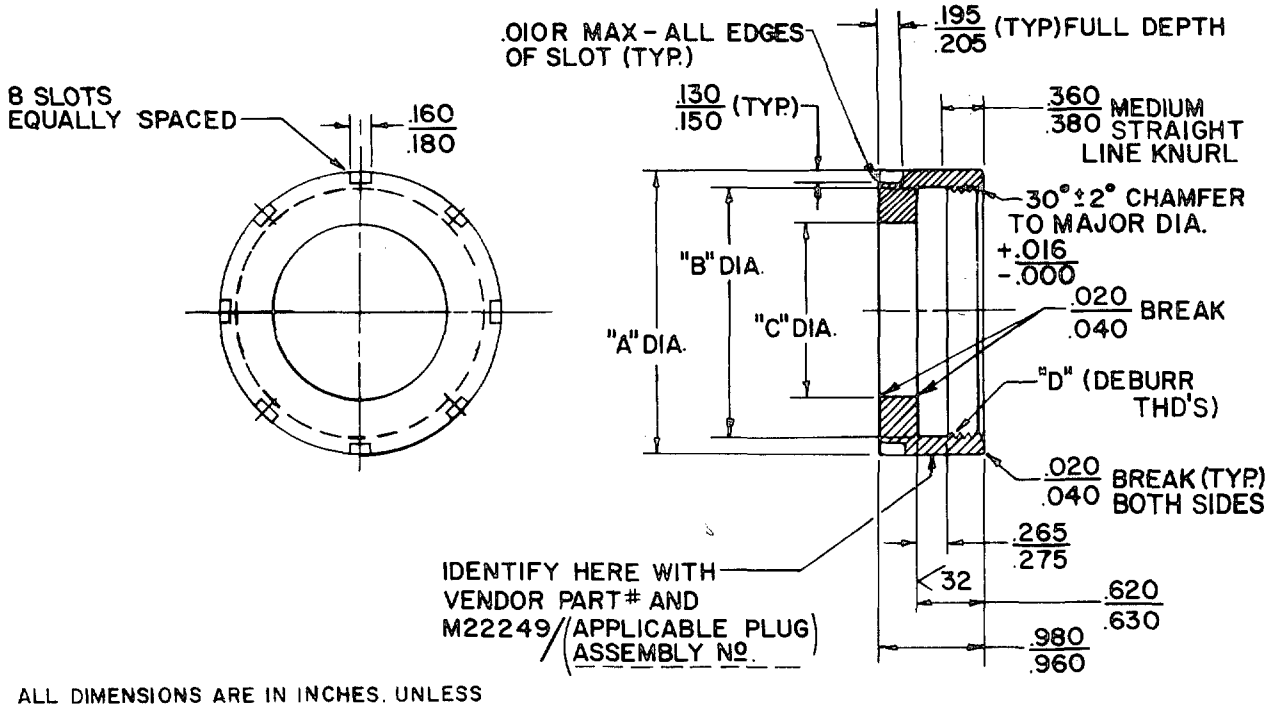


NOTE:
CONDUCTOR SEAL TO BE SHIPPED POSITION IN REAR INSULATOR AS SHOWN.
MATERIAL TO BE MOLDED SILICONE RUBBER.

ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE ± 1/64 INCH FRACTIONS AND ± 0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE SHALL BE ± 1/2 DEGREE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:39Z
Check the source to verify that this is the current version before use.

COUPLING JOINT
M22249/31-4



ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE ± 1/64 INCH FRACTIONS AND ± 0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE SHALL BE ± 1/2 DEGREE.

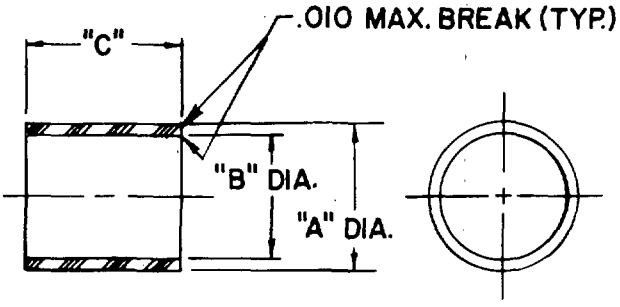
TABLE OF DIMENSIONS

	PART Nº	PIN SIZE & QUANTITY	DIA-A	DIA-B	DIA-C	THREAD-D
*	M22249/31-7	19 - #16	2.695/2.675	2.385/2.395	$1.657^{+.005}_{-.000}$	2-3/8-12UN-2B
*	M22249/31-6	12 - #16	2.385/2.365	2.185/2.195	$1.517^{+.005}_{-.000}$	2-1/8-12UN-2B
*	M22249/31-5	6 - #16	2.010/1.990	1.790/1.800	$1.203^{+.005}_{-.000}$	1-3/4-12UN-2B
	M22249/31-4	3 - #16	1.625/1.605	1.260/1.270	$.890^{+.005}_{-.000}$	1-1/4-12UNF-2B

*NOTE: NOT PART OF /31 ASSEMBLY

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:39Z
Check the source to verify that this is the current version before use.

REAR INSULATOR



ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE ± 1/64 INCH FRACTIONS AND ± 0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE SHALL BE ± 1/2 DEGREE.

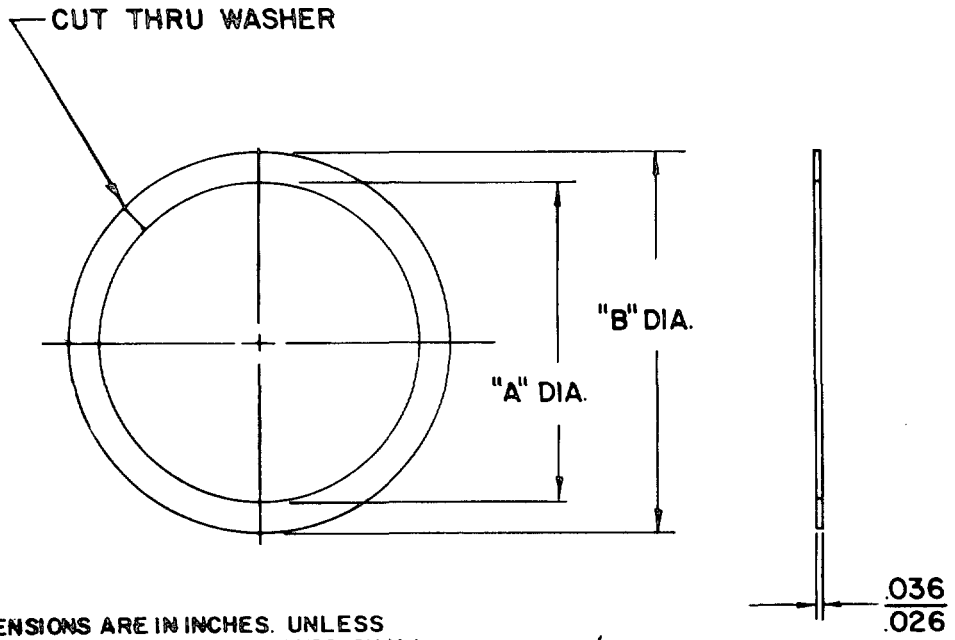
TABLE OF DIMENSIONS				
PART Nº	PIN SIZE & QUANTITY	DIA. "A"	DIA. "B"	DIA. "C"
* M22249/31-11	19-#16	1.256/1.246	1.080/1.090	.663/.658
* M22249/31-10	12-#16	1.132/1.122	.960/.970	.663/.658
* M22249/31-9	6-#16	.868/.861	.742/.749	.663/.658
M22249/31-8	3-#16	.600/.594	.437/.442	.369/.364

* NOTE: NOT PART OF /31 ASSEMBLY

Ⓐ MATERIAL: TO BE MOLDED FROM DIALLYL PHTHALATE IN ACCORDANCE WITH MIL-C-22249.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:39Z
Check the source to verify that this is the current version before use.

THRUST WASHER



ALL DIMENSIONS ARE IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES SHALL BE ± 1/64 INCH FRACTIONS AND ± 0.005 INCH FOR DECIMALS. ANGULAR TOLERANCE SHALL BE ± 1/2 DEGREE.

125/ ALL OVER

TABLE OF DIMENSIONS

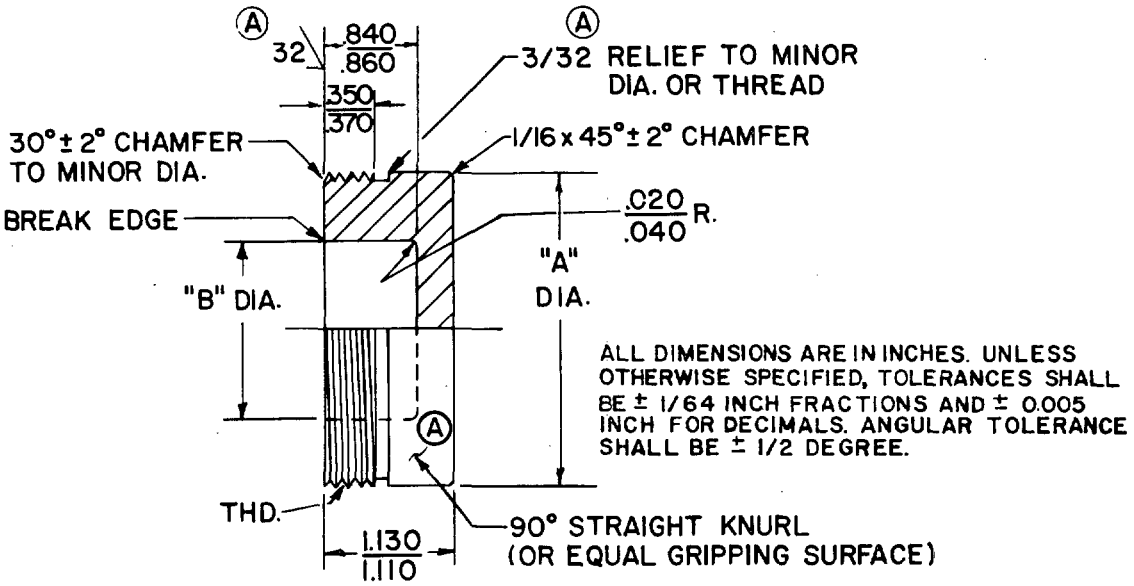
	PART Nº	PIN SIZE & QUANTITY	DIA. "A" (A)	DIA. "B"
*	M22249/31-15	19 -#16	1.657 $^{+.015}_{-.000}$	2.150/2.130
*	M22249/31-14	12 -#16	1.517 $^{+.015}_{-.000}$	2.010/1.990
*	M22249/31-13	6 -#16	1.203 $^{+.015}_{-.000}$	1.760/1.740
	M22249/31-12	3 -#16	.890 $^{+.015}_{-.000}$	1.260/1.240

* NOTE: NOT PART OF /31 ASSEMBLY

MATERIAL: NYLON

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:39Z
Check the source to verify that this is the current version before use.

PROTECTIVE COVER



Ⓐ NOTE: 125 MAX. FINISH EXCEPT AS NOTED.

TABLE OF DIMENSIONS

	PART Nº	PIN SIZE & QUANTITY	DIA. "A"	DIA. "B"	THD SIZE
*	M22249/31-19	19 -#16	2.377 / 2.357	1.560/1.578	2-3/8-12UN-2A
*	M22249/31-18	12 -#16	2.140/2.120	1.356/1.376	2-1/8-12UN-2A
*	M22249/31-17	6 -#16	1.760/1.740	1.091/1.103	1-3/4-12UN-2A
	M22249/31-16	3 -#16	1.260/1.240	.667/.687	1-1/4-12UNF-2A

* NOTE: NOT PART OF /31 ASSEMBLY

MATERIAL: STYRENE A.B.S. (CYCOLAC OR KRALASTIC) OR DELRIN IF MACHINED FROM BARSTOCK.
ALTN MAT'L: AL2024, T351 PER QQ-A-225/6 CLEAR ANODIZED PER MIL-A-8625.

Preparing activity:
Navy-SH
(Project 5935-N069-7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T11:39Z
Check the source to verify that this is the current version before use.